SUMMARY: Below is my work for NASA L'SPACE. As the Lead Systems Engineer, I worked with each part of the engineering team to create the requirements for the LunarLink rover. Following the table includes the high-level rover system design, along with a paragraph of design justification.

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Req met?
MECH-1	The mechanical system shall maintain structural stability under lunar conditions.	Required to withstand launch and lunar surface conditions.	SYS-1	-	Analysis	Met
MECH-2	The mechanical system shall be able to traverse lunar terrain while maintaining vehicular stability.	Required to safely traverse lunar terrain for science operations.	SYS-1	-	Analysis	Met
THRM-1	The thermal system shall have the ability to identify the change in thermal radiation exposure within the temperature range of chosen PSR.	STM Requirement	SYS-3	-	Test	Met
THRM-2	The thermal system shall be able to categorize surface temperatures as PSRs or not.	Required to meet STM	SYS-3	-	Analysis	Met
THRM-3	The thermal system shall measure the surface temperatures to indicate illumination and record the data for sunlight exposure.	Required to meet STM	SYS-3	-	Test	Met

THRM-4	The thermal system shall be able to characterize and record both surface and	Human exploration goal of STM	SYS-3	-	Test	Met

	near-surface temperatures to gather thermal gradients and extremes.					
ELEC-1	The electrical subsystem shall provide ample power to the payload components during scientific reconnaissance.	Required to power payload and system operations.	SYS-2	-	Analysis	Met
ELEC-2	The electrical subsystem shall provide enough power storage to support operational hours without solar energy support.	Required to operate without solar power in PSRs.	SYS-2	-	Analysis	Met
CDH-1	The CDH subsystem shall autonomously execute science sequences without real-time ground control.	PSR comms constraints.	SYS-5	-	Demonstration	Met
CDH-2	The CDH subsystem shall store science data until downlink is available.	Intermittent comms	SYS-2	-	Demonstration	Met
CDH-3	The CDH subsystem shall perform fault detection and recovery for critical operations.	Mission robustness.	SYS-5	-	Demonstration	Met
CDH-4	The CDH subsystem shall timestamp and tag science data with positional metadata.	STM traceability.	SYS-2	-	Demonstration	Met

PAY-1	The payload shall detect volatile indicators including hydrogen and ice proxies.	Required STM science objective.	MG-1	-	Demonstration	Met
PAY-2	The payload shall collect in-situ measurements from PSR regolith or surface.	Volatile characterization.	MG-1	-	Demonstration	Met
PAY-3	The payload shall measure surface temperature gradients within PSRs.	Environmental assessment.	MG-1	-	Demonstration	Met
PAY-4	The payload shall capture high-resolution surface imagery for terrain mapping.	ESDMD mapping goal.	SYS-4	-	Demonstration	Met
PAY-5	The payload shall operate within allocated mass, power, and data budgets.	System feasibility.	CC-1	-	Analysis	Met

Table 2.1.1: Spacecraft Overview Verification Table

Figure 2.1.2: Side view

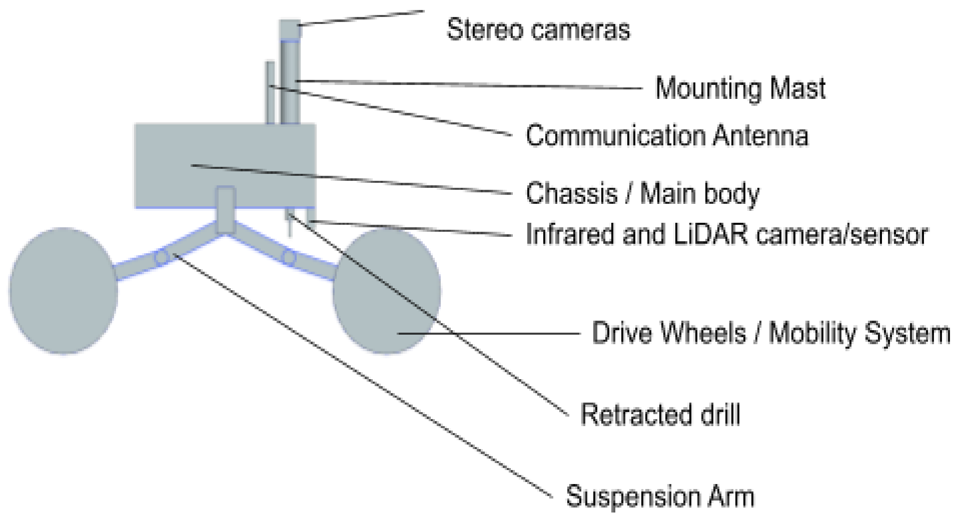


Figure 2.1.3: Isometric View

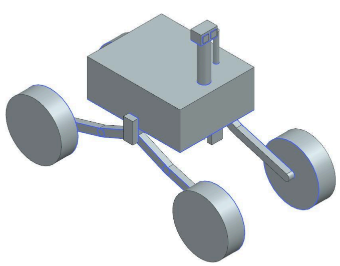


Figure 2.1.4: Top View

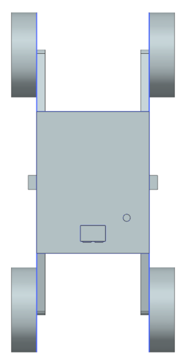
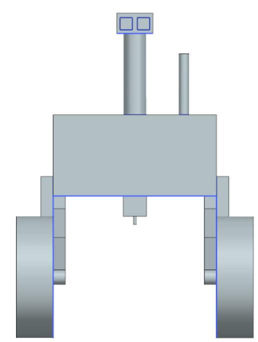


Figure 2.1.5: Front view



Subsystem	Component	Dimension	Value
Chassis	Chassis length	Length	500.0 mm
	Chassis width	Width	400.0 mm
	Chassis ground clearance	Height (chassis to floor)	345.0 mm
Mobility System	Tire width	Width	300.0 mm
	First linkage arm	Length (joint to joint ball)	184.3 mm
	Joint angle	Angle	172.5°
	Second linkage arm	Length (joint ball to wheel)	282.7 mm
	Joint ball	Width	37.5 mm
Communication System	Antenna	Diameter	25.2 mm
Sensor Mast	Mast pole	Diameter	55.4 mm
Stereo Camera Assembly	Camera housing	Length	88.4 mm
	Camera housing	Width	55.5 mm

	Stereo cameras (2)	Side length	30.0 mm
	Distance between cameras	Spacing	12.4 mm
Navigation Sensors	Infrared / LiDAR sensor	Length	57.0 mm
		Width	24.0 mm
Subsurface Instrument	Drill housing	Diameter	26.3 mm
		Height	30.0 mm
	Drill bit (retracted)	Diameter	8.7 mm

Table 2.1.6: Subsystem Values

Subsystem	**Estimated Mass**	**Estimated Max Power**
Mobility system	8.00 kg	45.0 Watts
Navigation sensors	4.00 kg	8.00 Watts
Communication system	0.100 kg	4.00 Wats
Science payload (drill + sensors)	50.3 kg	374 Watts
Total rover	62.4 kg	431 Watts

Table 2.1.7: Mass and Power Totals

This rover is a four-wheel mobile platform designed to traverse and conduct scientific measurements in the permanently shadowed region (PSR) of the moon. The rover will conduct on-site analysis of the lunar regolith and the surrounding environment to support the mission's scientific objectives, including subsurface thermal characterization, using the built-in drill on the undercarriage of the chassis. The system includes LiDAR and infrared sensors mounted on the undercarriage of the chassis, a mounted stereo camera on the top of the chassis as well as an antenna for communication. The rover architecture includes five primary subsystems: Mechanical, Power, Command and Data Handling (CDH), Thermal, and Science Instrumentation, the appropriate values for which are given in the following tables.

2.1.1. Mechanical Subsystem Overview

The mechanical subsystem is responsible for providing the structural integrity, mobility, and physical integration required for the LunarLink rover to successfully operate within a permanently shadowed region. At a system level, this subsystem consists of the primary chassis, mobility system, and structural mounting interfaces for payload and supporting subsystems. These elements are designed to work together to maintain stability, protect internal components, and enable controlled traversal across uneven and uncertain lunar terrain.

The chassis serves as the main load-bearing structure and provides mounting points for all subsystems, including payload instruments, power components, and thermal hardware. It is designed to withstand launch loads, landing impact forces, and operational stresses encountered during traversal. Material selection prioritizes high strength-to-weight ratio and thermal resilience to ensure performance within strict mass constraints while maintaining structural reliability in extreme lunar conditions. The structure also accounts for thermal expansion and contraction due to the wide temperature range experienced in PSRs.

The mobility subsystem enables the rover to traverse the lunar surface, including loose regolith, slopes, and rocky terrain. This includes wheel design, suspension and considerations, and drivetrain integration to ensure sufficient traction and stability. The